Zaid Khan

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EDUCATION

National Institute of Technology Srinagar Expected 2027
Bachelor of Technology

PROJECTS

ResearchComb | Python, Markdown, Research Workflows 2026

- Built an open-source skill bundle for OpenAI Codex and Google Antigravity (AI coding assistants), with twelve research workflows plus an update command covering literature search, comparison, drafting, critique, and citation checks.
- Designed a shared evidence record that separates source identity, access, and claim support; recorded Codex tests include 11/11 focused workflow cases and a 567-word comb-paper draft within the requested range.

AI-Powered Customer Support Agent | Python, NLP, BM25, Evaluation 2026

- Processed a 2.8M-record support dataset into 154,976 customer-agent pairs, with chronological conversation splits, duplicate removal, PII redaction, and leakage checks.
- Implemented an eight-class Naive Bayes intent classifier, top-five BM25 retrieval, evidence-grounded reply drafting, and confidence- and risk-based escalation guardrails.
- Built a reproducible evaluation harness across macro-F1, routing safety, automation coverage, escalation recall, and latency (95.95% escalation recall on 150 held-out cases).

Redis-Compatible In-Memory Server | C++17, TCP/IP, RESP, Multithreading 2025 – 2026

- Built a minimal Redis-compatible server with POSIX TCP sockets and RESP, supporting PING, SET, GET, DEL, KEYS, and SAVE.
- Handled concurrent connections with a thread-per-client model and mutex-protected unordered map storage.
- Added automatic snapshot persistence on writes and startup reload; verified live TCP round trips and recovery across restarts.

Standalone MOS 6502 CPU Core | C++17, CPU Emulation, Testing 2025 – 2026

- Completed an unfinished MOS 6502 CPU core with a 64 KB memory bus, instruction execution, and reset, IRQ, and NMI handling.
- Corrected binary and decimal-mode ADC/SBC, branch and page-crossing penalties, zero-page wrapping, stack operations, flags, BRK, and RTI.
- Added a runnable machine-code demo and 13 instruction-level tests covering arithmetic, branches, subroutines, stack behavior, and interrupts; all tests pass.

TECHNICAL SKILLS

Languages: C++, C, Python
Computer Science: Data Structures & Algorithms, Operating Systems, Computer Networks
AI/ML: Naive Bayes, BM25 Retrieval, Model Evaluation, Guardrails
Systems: POSIX Sockets, TCP/IP, RESP, Multithreading, CPU Emulation
Databases: MySQL, Redis
Technologies/Tools: Linux, Git, Make, pytest, ncurses

LEADERSHIP / EXTRACURRICULAR

Cursor Hackathon March 2026
Lead Organiser NIT Srinagar

- Led planning and execution of a two-day national-level hackathon in Kashmir, attracting 600+ students, developers, and technology enthusiasts.
- Coordinated logistics, participant operations, outreach, and on-ground execution across the two-day event.

Google Developer Student Club Spring 2023 – Autumn 2024
Operations Lead NIT Srinagar

- Organized workshops on Git, open-source contribution, and software development best practices.